

Project Description

Project Name: A to-do list organizer

Grading weight: This project accounts for 15% of the course mark and will be graded out of 100 marks (85 marks for the implementation and 15 marks for the report). A bonus up to 10 marks to be added to **the project mark** will be given for implementing at least **two** of the *bonus* features suggested below. Please do not be tempted to implement all the bonus features, as this will cost you too much time. Also note that these bonus marks will only affect the project mark.

Project Overview:

The goal of this project is to create a GUI based application that can store the to-do list of a user , display , organize, alter and alert user to coming events.

The program should:

- 1) Allow new users to register using user name , email , password..etc
- 2) Allow existing users to sign in and see their to-do list.
- 3) Allow users to enter a new task.
- 4) Allow users to delete an existing task.
- 5) Allow users to update a certain task.
- 6) Allow users to mark a certain task as “done”.
- 7) Allow users to mark a certain task as “pending/late”.
- 8) Allow users to choose to display all tasks in date order.
- 9) Make alert/reminder to user when an upcoming task is approaching.
- 10) The program should have a GUI implemented.
- 11) It should handle any input errors from user (such as wrong username password)..etc

Bonus Features (Optional): these are only examples of possible bonus features

1) **Productivity Analytics & Achievement Monitoring:**

The system should analyze user productivity using task completion statistics, delayed task tracking, performance trends, and graphical dashboards. It should generate productivity reports, efficiency scores, and AI-based recommendations to improve time management and work performance.

2) **AI-Based Intelligent Scheduling & Smart Work Planning:**

The system should use AI or intelligent algorithms to prioritize tasks, generate optimized schedules, detect deadline conflicts, and predict delayed tasks. It may integrate with calendar systems and provide smart reminders and workload management suggestions.

3) **Task Management Features:**

The system should support enterprise-level task management with features such as priority levels, deadlines, progress tracking, subtasks, team assignments, attachments, and task history. It should also include advanced search/filtering, Kanban boards, timeline visualization, recurring tasks, and intelligent task organization to simulate a professional productivity management platform.

Implementation language: Python will be used in this project. The resulting application should be graphical user interface (GUI) application. Java could also be accepted yet prior approval of module tutor would be required.

Team Size: 4 to 5 students

Important Plagiarism notice: You have to write your own code and your own report from scratch. Projects based on others code will receive a grade of zero in the entire project and report (even if the code is heavily re-factored/modified, etc.). Examples of such sources include (but is not limited to) code copied from the following sources: other teams, previous year projects, open-source software (or Internet in general), tutors, etc.

Project Deadline: Monday June 22nd 2026, 11:59 pm. Projects and reports should be submitted via blackboard (only one student from each team should submit the project). ***All Teams shall be required to have a meeting with instructor, to evaluate your work, any team/team member not attending the meeting will be granted a zero. Any member not able to answer questions during the discussion/viva will be penalized as follows:***

1 question missed= -10%

2 questions missed= -30%

3 questions missed = -50%

More than 4 questions missed =-90%

Project Report: In addition to your team member names, the report should include:

1. A brief description of your implementation including any bonus features included.
2. Any assumptions you adopted that are not mentioned in this description (if any).
3. A user guide including a usage example step-by-step with snapshots.
4. A list of programs (and associated data if any) you simulated. You should at least provide 2 programs. The programs must cover all features required from your system.
5. **A summary of how the work was split among your team members (who did what exactly)**
6. **Optionally, you can include a section about your experience working on this project. This section will NOT affect your grade in any way.**

What and how to submit: Submit a compressed folder (zip or rar) on the blackboard. The folder must contain: 1) All your code (the entire project as a zip file), 2) The files representing your test programs and data, and 3) Your report (**in PDF format , word files will cause loss of grades**)